CELLULAR ENDOMEMBRANE SYSTEM: ENDOPLASMIC RETICULUM AND GOLGI APPARATUS

INTRODUCTION

Eukaryotic cells, unlike prokaryotic, have a series of internal membranes that delimit specific membranous compartments with a particular morphology and function. This compartment network requires a complex transport system that allows the movement of molecules from one compartment to another. These membranous compartments (also called organelles), are the endoplasmic reticulum with its rough and smooth components, the Golgi apparatus and the lysosomes. The functions of these organelles, except for the smooth endoplasmic reticulum (SER), are related to the synthesis, processing, transport and localisation of a major part of the cellular proteins. The SER, on the other hand, plays a key role in the synthesis of lipids, and therefore it is related to providing the rest of the compartments with membranous supplies.

1. ROUGH EDOPLASMIC RETICULUM

1.1 Structure and composition

The rough endoplasmic reticulum (RER) consists of an interconnecting network of membranous flattened saccules (or cisternae) and tubules with ribosomes attached to them. These saccules are connected with the nuclear envelope and with the SER. In electron microscopy cuts the saccules appear as "profiles" (and we call its interior "lumen") which normally present a thickness of 20 to 30 nm, although sometimes they can appear dilated. (e.g. in plasma cells).

Claude, Porter and Fulham were the first, in 1945, to describe the RER using the electron microscope. Their work was later completed by Sjöstrand, Palade and Porter, who in addition, they related this organelle with protein synthesis.

The RER is present in all nucleated cells (except for spermatozoids) and is especially well developed in cells with a high protein synthesis rate. In an exocrine pancreatic cell, the RER can easily occupy 20% of the cell volume. cells with abundant RER present a characteristic called cytoplasmic basophilia, which consists in the affinity of basic dyes for certain regions of the cell cytoplasm. Cytoplasmic basophilia is seen in areas with abundant RNA from ribosomes, either found free in the cytosol or attached to the RER. Depending on the position of the basophilic component, three types of basophilia can be differentiated.

Diffuse basophilia, which implies that almost the whole cytoplasm is basophilic. We can normally see a non-basophilic region within the cell, which corresponds to the Golgi apparatus, and is normally called, Negative Golgi Region. An example of diffuse basophilia can be seen in plasma cells.

Basal basophilia: in this case, the basophilia is seen in the basal part of the cell. This is very common in secretory epithelial cells which have a large amount of RER in the basal cytoplasm. We find this basophilia in exocrine pancreatic epithelial cells and in the epithelial cells of the salivary glands.

Basophilia in clumps: the basophilic components appear in groups or clumps, which correspond to areas were the RER or the free ribosomes are accumulated within the cell. This basophilia is typical of hepatocytes and neurons (the basophilic areas in neurons correspond to the clumps or Nissl bodies).

Functional studies performed in the 60s of the last century demonstrated that the RER was involved in the synthesis and distribution of proteins. Using radioactive amino acids and autoradiography techniques in pancreatic acinar cells, it was found that amino acids introduced into a cell were incorporated into proteins, primarily in the RER. Longer incubation periods showed that the proteins had been transported to the Golgi apparatus, then to the secretory vesicles and finally secreted by the cell. Therefore, these experiments proved the existence a pathway for protein synthesis and secretion, which involved passing from the RER to the Golgi apparatus then to the cell membrane and finally to the exterior of the cell.

The RER membrane is a trilaminar structure typical to that seen in cell membranes, and its thickness is of 5-6 nm, in other words, a little thinner than the plasma membrane. Its composition is slightly different from other cell membranes as it it contains a higher percentage of lipids and a series of proteins exclusive to this organelle. Amongst these proteins, it is important to mention those involved in the attachment of ribosomes and the translocation of proteins towards the lumen. Moreover, the interior of the RER contains a series of proteins typical to this organelle, such as enzymes involved in the glycosylation of proteins, peptidases, disulphuric chains isomerases and chaperones.

1.2 RER protein translation

Cellular proteins can be completely translated by free ribosomes found in the cytosol or the translation may start in the cytosolic ribosomes and continue it in the RER. This difference is related to the destination of these proteins. Those synthesized in the RER will stay in this organelle, or will be transported to the Golgi apparatus, the lysosomes, the cell membrane or secreted to the extracellular space. On the other hand, those proteins translated by free cytosolic ribosomes can either remain in the cytosol or be posteriorly incorporated into the nucleus, mitochondria or peroxisomes. As the incorporation of these latter proteins into their target organelles takes place once the protein has been completely translated, this process is called post-translational importation. However, the ribosomes that join the RER membrane, translate the proteins at the same time as they are incorporated to the RER, and hence it is called co-translational importation.

The proteins that are going to be translated in the RER contain a signalling peptide of 16-30 residues found in the N-terminus region, as a result of the translation of a group of signalling codons present in the mRNA that codifies these proteins. The existence of peptide signals

in proteins, responsible for their incorporation in specific membranous compartments in the cell, was also proven in the case of the cell nucleus, mitochondria, chloroplasts and peroxisomes.

How does the translation and simultaneous incorporation of proteins into the RER take place? Several sequential steps can be distinguished:

Step 1: the translation starts in the free ribosomes. When the signalling peptide appears at the N-terminus end of the protein in formation, this peptide is recognized by the SRP (signal recognition particle) and this indicates that the protein must be translated in the RER. The SRP is made up of an RNA molecule and six proteins, and has several protein domains (for binding to the signal peptide, for stopping translation, for joining the ribosome and for the coupling of GTP.

Step 2: the binding of the SRP to the ribosome stops the translation and leads to a conformational change in the SRP which can now be recognized and bind to its receptor in the RER membrane. The ribosome/SRP complex and its receptor then interact with a translocator, a protein complex that will allow the protein that is being synthetized to pass into the RER lumen.

Step 3: the translocator is opened the protein synthesis continues. Then, the signal peptide remains anchored to the internal region of the translocator via protein interactions and hence will not pass into the RER. The protein being synthesized will work its way along the translocator and then penetrate the RER. The hydrolysis of GTP will take place resulting in conformational changes that will lead to the SRP release.

Step 4: a signal peptidase, bound to the inner surface of the membrane, cuts off the signal peptide from the protein that is being incorporated into the RER.

Step 5: translation continues and the protein will remain in the RER lumen.

1.3 Localization and orientation of the proteins in the RER

This mechanism of incorporation that we have just explained is not common to all proteins translated in the RER. there are proteins that remain joined to the membrane of this organelle because they have hydrophobic signals that allow them to stay in the lipid bilayer. Indeed, some proteins have, in addition to the signal peptide at the N-terminus domain, a stop-transfer sequence in an internal region of the protein. This sequence of 20 to 25 amino acids is hydrophobic, which prevents the protein from continuing its translocation towards the RER lumen. In this case, the protein translation continues, and the sequence that comes after the stop-transfer sequence will remain facing the cytoplasmic side. Therefore, once the translation finishes, we will obtain a protein anchored to the RER membrane, with its N-terminus end facing the RER lumen and the carboxyl end (C-terminus) facing the cytoplasm.

On the other hand, there are proteins that are orientated within the membrane in other ways, with the N-terminus end facing the cytosol, or with both ends facing the same side or even

there are proteins that cross the RER membrane several times. In order to achieve these orientations various strategies consisting of one or more sequences for initiating the translocation and stop-transfer sequences are needed.

Some proteins have the start transfer sequence located internally in the protein instead of at the N-terminus domain. In this case, the ribosome that translates it will only join the RER when this signal appears, which will determine that the N-terminus end will be facing the cytoplasmic side when its translocation through the RER membrane begins. The translation of the protein continues in such a way that the carboxyl end is placed facing the RER lumen. In this case, there is no removal of the signal sequence by the signal peptidase, thus leaving the protein embedded in the membrane with the hydrophobic signal immersed in it. The internal sequences of the protein can be arranged in such a way that they control the translocation orientating the N-terminus end towards the cytosolic side or the luminal side of the RER, which will in turn allow a different conformation of the protein in the membrane. By changing the direction of the start and stop transfer sequences proteins can be orientated with both N and C-terminus ends towards the same compartment (cytosolic or luminal side). The presence of several start and stop transfer sequences permits the existence of proteins that cross the RER membrane several times.

1.4 Post-translational modifications

The proteins translated in the RER must achieve their final configuration so as to be perfectly functional. This configuration depends upon the correct folding of the protein as well as a series of post-translational modifications that take place inside the RER (glycosylation, fragmentation and formation of disulfide bonds).

Among cellular proteins, only those that pass through the RER will be glycosylated, which it does not mean that all the proteins translated in the RER will undergo the glycosylation process (although most will). Glycosylation is a process that starts in the RER and ends in the Golgi apparatus. Both organelles contain enzymes capable of transferring sugars to the proteins or deleting some of them so as to form chains of specific oligosaccharides for each protein. Glycosylation is an important process because some proteins need to be glycosylated in order to fold correctly in the RER. Furthermore, glycosylation prevents protein aggregation.

All the proteins that are glycosylated in the RER first undergo a a transference in block of an oligosaccharide formed by 14 residues: 3 of glucose, 9 of mannose and 2 of N-acetylglucosamine, catalysed by an oligosaccharide transferase. The oligosaccharide has been previously synthetized on the cytosolic side of the RER membrane by means of the activity of several enzymes that build the sugar chain on a lipid in the membrane called dolichol-diphosphate. Once the 14 residue oligosaccharide has been made, a translocation of the carbohydrate chain takes place,, from the cytosolic side to the lumen RER. The transfer of the oligosaccharide to the protein will take place on an asparagine amino acid (Asn) that is found in a Asn-X-Ser/Thr sequence (where X can correspond to any amino acid; Ser:

serine; Thr: threonine). This oligosaccharide will be posteriorly modified in the Golgi apparatus.

The correct formation of the three-dimensional structure (or correct folding) of the proteins is a phenomenon of great importance. The formation of disulfide bridges (S-S) takes part in the process of acquisition of a correct protein structure. In proteins located in the RER lumen, disulfide bonds are formed by the loss of two electrons from the SH groups found in cysteine amino acids. This protein modification takes place in the RER because the oxidative environment favours the formation of S-S bonds, unlike the reductive environment found in the cytosol.

The folding of proteins in the RER is a complex process which is not completely understood. Chaperones and protein processing enzymes act as sensors, detecting misfolded proteins and sending them to the cytosol so that they can be degraded.

2. SMOOTH ENDOPLASMIC RETICULUM

The SER is formed by a system of membranous structures, mainly tubular, without attached ribosomes, and they differ from the RER in morphology, composition and function.

Although there are morphological and funcitonal differences between both reticula, they are interconnected with each other, forming a structure or compartment with a continuous membrane. The characteristics that differentiate the SER from the RER are:

- It does not have ribosomes.
- The main membranous elements are fundamentally tubules (not saccules).
- Its membrane composition it is not the same (different proteins and a greater percentage of lipids).
- Its functions are mainly related to the synthesis of lipids and fat derivatives.
- It is only abundant in specialized cells such as hepatocytes, steroid hormone secretory cells, muscular cells, etc.

The SER has numerous functions within the cell, mainly related to the synthesis of lipids, which are necessary for the increase and replacement of the membranous elements of the different organelles and the plasma membrane.

2.1 Functions

LIPID AND FAT DERIVATIVES SYNTHESIS

In every cell the synthesis of phospholipids and cholesterol is very important for the renewal of cellular membranes. This is done by the SER. Moreover, there are specialised cells in the organism in which the SER is very abundant; in these cells the synthesis of certain types of lipids and their derivatives takes place.

The lipid synthesis takes place in the cytosolic hemimembrane of the endoplasmic reticulum, therefore, the addition of lipid molecules would only take place on this side of the membrane.

In order to redistribute the lipids in both hemimembranes and establish the asymmetry that characterizes them regarding their composition, the activity of enzymes called scramblases is required, which catalyse the energetically unfavourable passage of the lipid polar groups from one side of the membrane to the other (see chapter 3).

The lipids synthesized in the SER will be distributed by means of vesicular transport to the different membranous compartments (RER, Golgi apparatus, lysosomes) and to the plasma membrane. In order for them to be transported to mitochondria and peroxisomes cytosolic proteins that exchange phospholipids are required

Due to its role in lipid synthesis, the SER also contributes in a very important manner in the formation of lipoproteins, bile acids and steroid hormones. Enterocytes contain RER and SER that take part in the formation of chylomicrons which are formed by triglycerides, phosphoglycerides, cholesterol and proteins. The liver also synthesizes lipoproteins (LDL, VLDL, HDL and VHDL, which are posteriorly released into the bloodstream) and bile acids, formed from cholesterol (and excreted into the bile canaliculi). The steroid hormone synthesis from cholesterol, takes place in specialized endocrine cells in several successive steps that require the action of mitochondrial and SER enzymes. In this way, hormones such as testosterone, oestradiol, cortisol and aldosterone etc. are synthetized.

DETOXIFICATION OF LIPOSOLUBLE SUBTANCES

The SER plays an important role in the breakdown of toxic liposoluble substances, such as certain drugs, alcohol, pesticides and carcinogens, as well as toxic substances produced by the metabolism itself. All these substances are turned into hydro soluble products, which are more easily eliminated by the organism. In hepatocytes, when there are high levels of such toxic substances, the SER is highly developed and contains detoxifying enzymes (such as the ones of the cytochrome P450 family, which are anchored to the cytosolic side of the SER membrane. The activity of these enzymes include oxidation, hydrolysis and reduction of substances which have to be chemically modified to reduce their toxicity. Other cytoplasmic enzymes act by conjugating these substances to certain chemical groups (glucuronic, sulfate, acetyl, etc), facilitating their excretion.

PARTICIPATION IN THE GLYCOGEN METABOLIM

In some cell types, the SER is frequently found close to glycogen stores, especially in hepatocytes. These cells play an important role in the storage of glucose as glycogen. On the cytosolic side of the SER membrane the glucose-6-phosphatase is present, an enzyme that hydrolases glucose-6-phosphate, producing free glucose that can go to the bloodstream and then to the tissues when their glucose levels are low.

Ca2+ STORAGE

As explained in chapter 4, the SER has Ca2+ pumps, Ca2+ ATPases that introduce Ca2+ into the SER, against the concentration gradient with the purpose of maintaining low cytoplasmic levels of this ion. ; therefore the SER acts as an intracellular reservoir of Ca2+. The increase in the cytoplasmic Ca2+ levels triggers signaling cascades that can have different consequences, depending on the cellular type (see chapter 13). Therefore, it acts

as a second messenger in intracellular signaling. In skeletal muscle fibers, a special type of SER is found, called sarcoplasmic reticulum, which wraps the myofibrils. It constitutes a Ca2+ storage. These ions are released with the arrival of the nervous impulse and then muscle contraction is produced.

3. GOLGI APPARATUS

3.1 Structure and properties

The Golgi apparatus is formed by a group of membranous structures, mainly flattened saccules and stacked in a parallel fashion. Vesicles also appear around the saccules, especially near its ends, as well as tubular shaped membranous elements. All these elements are grouped together forming dictyosomes, whose number and distribution varies according to the cell type (e.g. hepatocytes can have up to 50 dictyosomes). The Golgi apparatus is highly developed in cells that produce glycoproteins and proteoglycans, such as the pancreatic acinar cells, goblet cells, hepatocytes, neurons, etc.

The Golgi apparatus exhibits polarity, which means there are two areas with very different morphology and composition, called Cis face (or entry face) and Trans face (or exit face). Protein maturation takes place in the Golgi apparatus that occurs progressively from the Cis face to the Trans face. Hence protein maturation is polarised towards the trans face of the Golgi apparatus. For this, there is a clear functional compartmentalisation within the Golgi apparatus in such a way that it is possible to distinguish five compartments

- Cis Golgi network (CGN)
- Cis cisternae
- Medial cisternae
- Trans cisternae
- Trans Golgi network (TGN)

The two end compartments (CGN and TGN) are not formed by cisternae but by an irregular network of tubules and vesicles (this explains why they are called 'networks'). Each compartment has a specific group of enzymes responsible for the sequential maturation of the proteins. Therefore, in the CGN region and in the Cis area we can find enzymes that transfer phosphate groups and eliminate mannose residues, respectively. In the medial cisternae N-acetylglucosamine is added; in the Trans ones, galactose and sialic acid, whilst in the TGN area enzymes transfer sulphate groups to certain proteins.

3.2 Functions

In the Golgi apparatus the proteins and the lipids synthesized in the ER will continue their maturation. The main changes that they will undergo are N-glycosylation, O-glycosylation, fragmentation and sulphation.

GLYCOSYLATION

Regarding the proteins that have been previously glycosylated in the RER (where the N-glycosylation took place), in the Golgi apparatus they will lose some of their sugar radicals and have new ones added, giving rise to glycoproteins which contain N-oligosaccharides that can be classified into 3 groups a) rich in mannose, b) complex and c) hybrids, (intermediate oligosaccharides, in between these two groups).

Apart from the modification of the oligosaccharides of the N-glycoproteins, the Golgi apparatus is also responsible for the O-glycosylation (which takes place exclusively in this organelle). In this type of glycosylation sugars are linked to an OH group of the following amino acids: serine, threonine and, in the case of collagen, hydroxylysine. The O-glycosylated proteins can be found forming mucins, which are abundant in cells that secrete mucinous substances; collagen also has O-ligated sugars.

MARKING OF LYSOSOMAL ENZYMES

Enzymes of the acid hydrolase type, which are going to be responsible for the cellular digestion that is carried out by lysosomes, are classified by means of a special route that will direct them towards this organelle. Primary lysosomes are formed from clathrin-coated vesicles, that originate in the TGN area of the Golgi apparatus. The process begins in the CGN area, which contains enzymes that add phosphate groups to the 6th carbon of the terminal mannose on the hydrolase proteins, that will go to lysosomes. Mannose-6-phosphate (M6P) therefore constitutes a way of tagging those proteins that will go to the lysosome. All these proteins undergo N-glycosylation as well as the addition of a phosphate group which takes place as a result of the enzyme activity of N-acetylglucosamine phosphotransferase. This enzyme transfers a phosphate group from N-acetylglucosamine-UDP to the mannose found in lysosomal enzymes. The labelled lysosomal hydrolases are recognized by M6P receptors found in the TGN area. In this way, clathrin coated vesicles are formed which leave the TGN area and fuse with the late endosome. Any alteration in this classifying system will produce serious consequences for the cell, since these enzymes are not destined for the lysosome, but they join the secretory pathway and are released into the extracellular space.

MEMBRANE RECYCLING

The Golgi apparatus is a very dynamic compartment that must be in constant equilibrium, as it continuously receiving contributions of membranous elements that come from endocytosis, and in turn releases vesicles or the secretory pathway, whose membrane becomes part of the plasma membrane. It is therefore an organelle responsible for the correct distribution of the different membranous components of the cell.

PROPER DISTRIBUTION OF THE DIFFERENT MACROMOLECULES

As we have already mentioned, the Golgi apparatus is a classifying compartment, from which vesicles emerge with different cell destinations. Proteins can be directed towards the plasma membrane and stay there (if they are proteins anchored to the membrane), be released to the cell exterior, stay as Golgi resident proteins or be destined for the lysosomes.